

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2017

Math 1117
(Mathematics)

Full Marks: 210

Time: 3.00hrs

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define invariants. Find the transformed equation of $3x^2 + 2xy + 3y^2 = 0$ when the origin is shifted at (2, 3) and the axis are rotated through an angle 45° . (12)
- (b) Describe a set of relation to transfer Cartesian coordinates system to spherical polar coordinate system. Here find the spherical polar coordinates of the point whose Cartesian coordinates are (0, -2, $-\sqrt{2}$). (17)
- (c) Write down the condition for parallelism of a line and a plane. Under what condition a plane will contain a line? (06)
2. (a) Identify the conic $6x^2 + 5xy - 6y^2 - 4x + 7y + 11 = 0$. Also find its centre if it is a central conic. (11)
- (b) A plane meets the coordinate axis in A, B, C such that the centroid of the triangle ΔABC is the point (a, b, c). Prove the equation of the plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3$ (11)
- (c) Examine the four points (0, 1, 2), (3, 0, 1), (4, 3, 6) and (2, 3, 2) are coplaner or not. If they are non-coplaner, then find the value of the tetrahedron, where the verticals are those four points. (13)
3. (a) Find the distance of the point (1, -1, 1) from the line: $2x + 3y + 4z - 5 = 0 = 3x - 2y + z + 4$. (12)
- (b) In the spherical triangle ABC, $\angle C = 90^\circ$, $a = 134^\circ 30'$, $c = 122^\circ 8'$, find the values of $\angle B$ and b. (11)
- (c) Find the length of shortest distance between two lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$. (12)
4. (a) Define spherical triangle. Derive the fundamental relation between sides and angle of a spherical triangle ABC, $\cos a = \cos b \cos c + \sin b \sin c \cos A$. (14)
- (b) Define direction cosine of a line. Prove that $l^2 + m^2 + n^2 = 1$, where l, m, n are direction cosines of a line. (10)
- (c) Find the great circle distance between two places, Dhaka ($23^\circ 44' N$, $90^\circ 24' E$) and Karachi ($24^\circ 52' N$, $67^\circ 1' E$). (11)

Section – B

5. (a) Discuss with example those arithmetic operation and rules in matrices which are different from that of operations and rules in real numbers. (13)

- (b) Express $A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 5 & 7 \\ 3 & 2 & 9 \end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrices. (12)

- (c) Prove that $B \cdot \text{adj } B = |B| \cdot I$. Here find $|B \cdot \text{adj } B|$, where B be any square matrix. (10)

6. (a) Define the following: (09)

- i. Upper triangle matrix
- ii. Periodic matrix
- iii. Symmetric matrix

- (b) Reduce the following matrix of its echelon form, canonical form and finally (16)

normal form. Hence find its rank, where $A = \begin{bmatrix} 0 & -4 & 1 & -2 \\ 1 & 2 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{bmatrix}$

- (c) If each column of the matrix A be the vectors X , Y and Z respectively, check (10)

whether vectors are linearly dependent or linearly independent. If possible express the vectors as a linear combination; where $A = \begin{bmatrix} 0 & -4 & 2 \\ -2 & 2 & 2 \\ 2 & 0 & -1 \end{bmatrix}$

7. (a) (14)

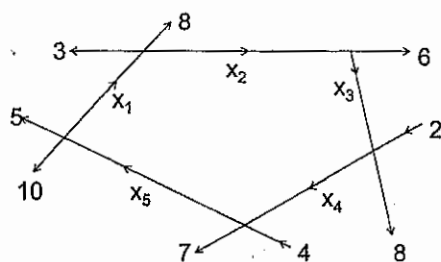
If $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$ find A^{-1} . Now given that $AX=H$, $H = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$; Then find the value of X .

- (b) Test the consistency of the following system of liner equations. If possible find (15)

the solution of the system:
$$\begin{aligned} 3x + 2y + 3z &= 16 \\ 6x - 3y + 5z &= 15 \\ 9x + 4y - 7z &= -4 \end{aligned}$$

- (c) Test where the matrix $A = \frac{1}{3} \begin{bmatrix} -2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{bmatrix}$ is orthogonal or not. (06)

8. (a) Mathematically formulated the following network flow diagram. Hence (08)



- (b) Draw e^x and e^{-x} in a graph paper. Now graphically show the relation among (20)

general hyperbolic function, i.e. $\sinh x$, $\cosh x$ and $\tanh x$.

- (c) Define logistic function curve (including meaning of parameters) (07)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2017
URP 1111
(History/Evolution of Human Settlements)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What are the elements of human settlements? Explain with relevant examples. (12)
(b) Imagine you were alive hundreds of years ago and you had to choose a site for your settlement. What site selection factors for human settlements would be important to you? (13)
(c) Draw the hierarchy pyramid of urban settlements according to population. (10)
2. (a) Write a short note on Ziggurats in Mesopotamian civilization with necessary sketches. (13)
(b) Briefly explain city planning concepts of Mesopotamian civilization with necessary sketches. (12)
(c) Nile River is known as the "Giver of Life" during Egyptian civilization. Clarify the statement with relevant examples. (10)
3. (a) Parthenon, the most important and characteristic monument of ancient Greek civilization still remains its international symbol. Discuss with necessary sketches. (15)
(b) Discuss the grid-iron patterns of streets during Greek period with sketches. (08)
(c) Explain the structure and governments of city states in Greek civilization with examples. (12)
4. (a) 'Pantheon' is the magnificent ancient temple in Rome. Discuss with necessary sketches. (15)
(b) Discuss about Roman construction techniques with sketches. (08)
(c) Briefly describe 'Mohenjo-Daro' and 'Harappa' city life. (12)

Section – B

5. (a) Briefly describe the changes from Old to New Stone Age. (10)
(b) Mention the major innovations of the Stone Age. (05)
(c) "Farming did not start until the New Stone Age". Why and what led to this development? (10)
(d) Describe the various stages in the evolution of human settlement. (10)
 6. (a) Differentiate between the following: (10)
 i) Primitive non-organized and Primitive organized
 ii) Static urban and Dynamic urban settlements
(b) Describe Beam-framed house. (10)
(c) How did the houses develop in Iron Age II (10)
(d) Give examples of Solomonic fortifications. (05)
 7. (a) Briefly describe the various towns according to their functional classifications. (15)
(b) Describe the factors that facilitate Rural-Urban Migration. What are its consequences? (10)
(c) Write short notes on the ancient settlements of: (10)
 i) Vanga ii) Bagherhat, Khulna
 8. (a) Briefly describe the vertical city of Le Corbusier. (20)
(b) Briefly describe the three examples of Garden City. (15)
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Department of Urban & Regional Planning
BURP 1st Year 1st Term Regular Examination, 2017.

Ch 1117
(Environmental Chemistry)

Time: 3 hrs

Full Marks: 210

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section - A

1. (a) What is mass defect? How it is related to the binding energy and stability of the nucleus? 10
(b) Explain about radioactive decay. Describe the group displacement law of radioactive decay. 10
(c) What is nuclear reactor? How electricity can be generated with the help of nuclear reactor? 10
(d) A sample of carbon from a wooden artifact is found to give 7.00 $^{14}_6\text{C}$ counts per minute per gram of carbon. What is the approximate age of the artifact? The $^{14}_6\text{C}$ from recently cut down wood decays at the rate of 15.3 disintegrations per minute per gram of carbon and half life of $^{14}_6\text{C}$ is 5770 years. 05
2. (a) What is ionization energy? How does ionization energy changes from Li to Cs? Explain reason. 10
(b) "Transition metal form color compound" – explain. 10
(c) "Transition metals are suitable for using as catalyst" – why? 08
(d) "Alkali metals are more reactive than corresponding alkaline earth metals" – explain. 07
3. (a) Describe Valence Bond Theory (VBT) for H_2 molecule. 13
(b) "Oxygen molecule (O_2) is paramagnetic" – explain with the help of molecular orbital theory. 10
(c) Describe H- bonding and its application. 12
4. (a) Explain about the colligated properties of solution. 07
(b) What is lowering of vapour pressure of a solution? Derive a relation between lowering of vapour pressure and mole – fraction of solute and solvent. 10
(c) State and explain the Henry's law of gas solubilities with its limitations. 10
(d) "HCL is a Bronsted acid cannot be called a Lewis Acid" – Shortly mention the requirements of Bronsted acid and Lewis acid. 08

Section - B

5. (a) Define electrophile and nucleophile with suitable example. Is chlorine free radical ($\text{Cl}^\bullet$) electrophile or nucleophile? Explain. 09
- (b) Write short note on the followings : 09
- (i) Inductive effect
- (ii) Resonance effect.
- (c) What is living polymer? Discuss ionic polymerization process with mechanism. 12
- (d) Write down the differences between thermosetting and thermoplastic polymer with suitable example. 05
6. (a) Draw and explain the major region of the atmosphere with temperature profile and also write the altitude and important chemical species in each region. 12
- (b) Explain the Nitrogen Cycle of an environment. 10
- (c) What is green house effect? Describe causes and adverse effect of green house effect. 08
- (d) What do you mean by volatile organic compounds (VOC_s)? Discuss about the toxicity of VOC_s . 05
7. (a) Define pollution and pollutant. Write down the sources and effect of the following pollutants : 13
- (i) CO_2 and CO
- (ii) NO and N_2O
- (iii) Hydrocarbons.
- (b) Briefly illustrate the different types of impurities that are usually found in natural water. 10
- (c) What do you mean by effluent treatment plant? Discuss about the effluent treatment plant of waste water. 12
8. (a) What is photochemical smog? How is smog formed? 10
- (b) What is threshold value? Describe the effect of different factors on atmospheric corrosion with the help of a diagram. 10
- (c) Write down the anode and cathode reaction in under-water corrosion. Clarify the atmosphere on the basis of differences in corrosion. 07
- (d) Discuss about anodic protection of corrosion. 08

Khulna University of Engineering & Technology
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BURP 1st Year 1st Term Regular Examination, 2017

URP 1113
(Fundamentals of Planning Process)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. Annotate any five of the following: (35)
 - (a) Project
 - (b) Programme
 - (c) Strategic Plan
 - (d) Structure Plan
 - (e) Detailed Area Plan
 - (f) Advocacy Plan
 - (g) Land Use Plan
2.
 - (a) "Urban Planning is related to three key aspects (social, economic and physical environment)" - Briefly explain the statement with proper example. (13)
 - (b) "Urban planning has three dimensions in terms of Space, Sector and Time". Briefly explain with relevant examples in the context of Bangladesh. (12)
 - (c) What is understood by "Urban Area"? What are the conditions to be an urban area (Municipality) as per the Pourashava Ordinance 2009? (10)
3.
 - (a) Draw the cyclic process of planning with several phases. (06)
 - (b) Briefly explain the terms Vision, Mission, Goal, and Objectives with relevant examples especially in context of Urban and Regional Planning. (12)
 - (c) Differentiate Structure Plan and Master Plan with appropriate examples. (10)
 - (d) Draw a schematic diagram of four stages plan for a city (a city is considered as a local scale). (07)
4.
 - (a) The lifecycle of urban development contains several phases. Briefly explain the phases with pertinent examples. (12)
 - (b) Write down the structure of Planning Commission. (08)
 - (c) Planning Commission has a three-fold role in socio-economic development of the country - Explain with example. (07)
 - (d) "There are different types of institutional linkages of Planning Commission with other Ministries/Divisions/Organizations" - write down these linkages. (08)

Section – B

5. (a) Discuss the concept of "Planning". "Planning is more needed in the developing country like Bangladesh" – Explain. (10)
- (b) "Planners' roles have changed in different ways from the beginning of planning till now" – Provide a brief overview of that scenario. (15)
- (c) "A separate town planning profession gradually developed after the first world war" – Write about the initiatives that contributed the most here. (10)
6. (a) List the five basic requirements of a Comprehensive Plan as mentioned by Goodman. (05)
- (b) Comprehensive Plan is also known as Land Use Plan. Explain with example. (15)
- (c) Mention some criticisms of the Rational Planning Theory. (05)
- (d) Make a comparison between "Positivism" and "Post-positivism" concept of planning theory. (10)
7. (a) Explain the Mixed Scanning Approach of Planning with the help of an appropriate example. (10)
- (b) Provide a tabular format of comparison of 'steps' and 'principles' between Rational Planning and Incremental Planning. (10)
- (c) What do you understand by "Developing a Communication Strategy" for an Advocacy Planning Process? (10)
- (d) Briefly discuss the concept of inclusive city. (05)
8. (a) Explain the "SITAR" Model. How has the SITAR Model derived its name? (10)
- (b) Write short note on any five of the following: (25)
- (i) Top-down and Bottom-up Approach (Comparison)
 - (ii) The '3E's of Planning
 - (iii) Radical Planning
 - (iv) Spatial Planning
 - (v) Factory Towns
 - (vi) Limitations and Strengths of Advocacy Planning
 - (vii) Components of Transactive Planning

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Examination, 2017
Hum 1117
(Functional English)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Make sentences using the following Modals as directed: (14)
 - i. Could (to express polite request)
 - ii. Could (to express opportunity in the past which was not executed)
 - iii. Had better (to express preference)
 - iv. May (to express guess about the future)
 - v. Should (to express duty in the present)
 - vi. Must (to express internal obligation)
 - vii. Need (to express necessary action in the past)
- (b) Make sentences on the following structures. (12)
 - i. Subj. + Intransitive verb + preposition + noun/pronoun
 - ii. Subj. + Linking verb + adjective complement
 - iii. Subj. + Transitive verb + direct obj. + adverbial phrase
 - iv. Subj. + Transitive verb + pronoun + what + infinitive
 - v. Subj. + Transitive verb + object + adj. complement.
 - vi. Subj. + Linking verb + noun complement + extension
- (c) Make sentences using the following phrases and idioms. (09)

Weal and woe ; Out and out ; Stone throw ; Ins and out ; Burst into ; Break down.
2. (a) Make new words with each of the following prefixes and suffixes and use them in sentences. (14)

Fore____, Mal____, Pro____, Super____, ____en, ____hood
- (b) Make sentences expressing the following emotions/notions. (12)
 - i. Threat
 - ii. Imprecation
 - iii. Invitation
 - iv. Annoyance
 - v. Apology
 - vi. Condolence
- (c) Identify the parts of speech of the underlined words of the following sentences. (09)
 - i. Our blessings come from above.
 - ii. Analyze the above sentence.
 - iii. The moral law is above the civil.
 - iv. He lost his all in speculation.
 - v. All that glitters is not gold.
 - vi. He was all alone when I saw him.
3. (a) Frame WH questions from the underlined parts of the following answers. (14)
 - i. We read for pleasure
 - ii. The teacher rebuked the student
 - iii. He has had his meal in a hotel
 - iv. He visited London, Paris, Berlin.
 - v. Rajshahi is famous for mango.
 - vi. The shop there is mine.
 - vii. He has been reading for three hours.
- (b) Make sentences using the following words as directed (12)

Man (as verb) ; School (as verb) ; School (as adjective) ; Study (as verb) ; Study (as adjective) ; Book (as verb)
- (c) Change the following words as directed and make sentences with the changed words (09)

Man (into adjective) ; Study (into adjective) ; Book (into adjective) ; People (into verb) ; People (into verb) ; People (into adjective) ; People (into noun)
4. (a) Transform the sentences as directed (14)
 - i. I wish to know the time when he departed. (Simple)
 - ii. He desires to succeed therefore he works hard. (Complex)

- iii. As you have already decided, you need not ask me anything about it. (Interrogative)
- iv. Why did he finish it? (Passive)
- v. What a nice moment! (Assertive)
- vi. She is one of the most regular students. (Comparative)
- vii. I like him the most. (Positive)
- (b) Complete the sentences with subordinate clauses. (12)
- i. _____ is known to all. (Noun clause)
- ii. I wonder _____ (Noun clause)
- iii. I know the place _____ (Adjective clause)
- iv. I live in a house _____ (Adjective clause)
- v. Strike _____ (Adverb clause of time)
- vi. Walk slowly _____ (Adverb clause of purpose)
- (c) Correct the following sentences (09)
- i. It is raining cats and dogs for two hours
- ii. He has come home yesterday.
- iii. Tell me where does he live?
- iv. Trust to God.
- v. He needs not go there.
- vi. Either of this three pen will do.

Section – B

5. (a) Read the following passage carefully and answer the questions that follow: (20)
- It is easy to destroy, but difficult to build a grand and beautiful thing. It takes a very long time to produce something grand and beautiful. You can look at the Jamuna Multipurpose Bridge. It takes many years, much labor, much money, etc. to build this great bridge, by which we succeed many things easily, but we couldn't do before. Also you can look at the Jamuna. Thousands of years take to create this great river, where we observe many scenes- wave, different types of fish, human beings' life thereon, ships, etc. Besides, you can look at the invention of atom. Much and invaluable research is made on this invention and it does a great help to human beings. Even your life, if you think it to make excellent, takes a long and steady perseverance to reach so grandeur. Not a single day and a single effort will make it so grand and beautiful. So, much patient perseverance in a long time can create a beautiful state of life though it is possible to destroy it within short time. Therefore, after we know of the hard way to build a beautiful entity, we should take care of it instead of using the thing in an abusive way. History too can make us aware of the fact. You can know the city of Rome. It was not built in a day. In the very beginning it was a small village of huts. By and by it began to progress. It became the center of art, literature, culture, civilization, etc. This is true of everything of every great achievement. It is not the result of one day and little effort and perseverance. It achieves greatness only after years of patient labor. We want to be at once rewarded for our labor. This is not possible. There is no short-cut to do great things. Every great achievement requires continuous and hard labor. Our pace should be slow but sure. Only then we can do something great. The Pyramids of Egypt and the Taj of Agra are wonderful buildings. They were not built in a day. They are the result of very hard work and sacrifice. Thus, to do great work, we must learn to work hard and patiently. In addition we should take care of it. Otherwise civilization goes astray.
- i. What does one need to create a beautiful and grand thing?
- ii. What should we do to preserve a beautiful thing? And why?
- iii. What shape of beauty does the city of Rome take and how does it become so?
- iv. When can we do great things? Exemplify
- (b) Make a précis of the above written passage (Q.5.a) with a suitable title. (15)
6. (a) Write a listing paragraph on "Load Shedding" (Around 1000 word). (20)
- (b) Prepare a report on your dining of the varsity (Around 800 words). (15)
7. (a) Amplify the idea "Uneasy lies the head that wears a crown". (Around 100 words). (20)
- (b) Write a letter to your younger brother on the dignity of a man in humanities, not in matter (Around 800 words). (15)
8. Write a free composition on one of the followings: (Around 1800 words). (35)
- (a) The importance of reading in intellectuality building
- (b) Preserving our environment.